

Topic 8: IIR Filters

[Summary based on J. H. McClellan, R. W. Schafer, and M. A. Yoder, *Signal Processing First*, Prentice Hall, 2003.]

Topics: recursive difference equations, initial-rest conditions, impulse and step responses, system functions, direct-form structures, regions of convergence, poles and zeros, stability, frequency response, inverse z-transforms, transient and steady-state behaviour, and second-order IIR filters.

Key idea. IIR filters use feedback, so a finite input can produce an infinite-duration response. Their behaviour is governed compactly by the poles and zeros of a rational system function: pole locations determine stability and transient decay, while evaluating the system function on the unit circle gives the frequency response.

8.1 The general IIR difference equation

The most general class of implementable digital filters is obtained when the output is computed not only from the input but also from previously computed outputs. The equation is

$$y[n] = \sum_{l=1}^N a_l y[n-l] + \sum_{k=0}^M b_k x[n-k]$$

When the coefficients are zero, the difference equation reduces to the FIR-filter difference equation. $\{a_l\}$

The difference equation of a first-order IIR filter is

$$y[n] = a_1 y[n-1] + b_0 x[n] + b_1 x[n-1]$$

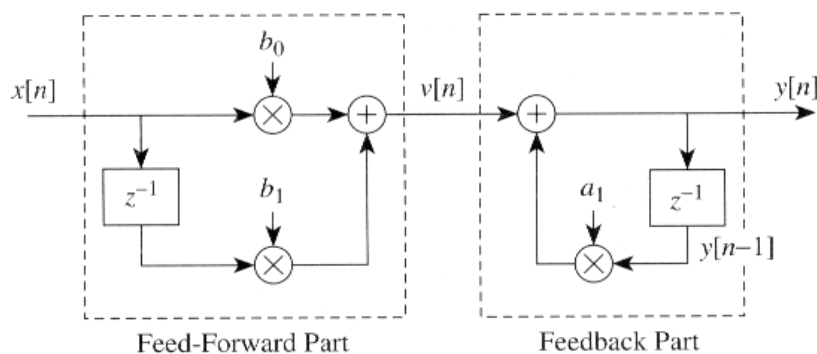


Figure 8.1. First-order IIR system showing one recursive coefficient and two non-recursive coefficients.

8.2 Time response

To understand IIR filters, consider the following example:

$$y[n] = 0.8y[n-1] + 5x[n]$$

with the input signal

$$x[n] = 2\delta[n] - 3\delta[n - 1] + 2\delta[n - 3]$$

Assume that the input signal is zero for $n < 0$. Therefore, evaluating the output at $n=0$ gives

$$y[0] = 0.8y[0 - 1] + 5x[0] = 0.8y[-1] + 5(2)$$

The problem is immediately apparent: the value $y[n]$ at $n=-1$ is unknown. The solution requires the following assumptions, called initial-rest conditions:

1. The input must be zero before the starting time; that is, $x[n]=0$ for $n < 0$. $n_0 \ x[n] = 0 \ n < n_0$
2. The output must also be zero before the starting time of the input signal; that is, $y[n]=0$ for $n < 0$. $y[n] = 0 \ n < n_0$

Using these conditions, we can evaluate the output of the preceding system:

$$y[0] = 0.8y[0 - 1] + 5x[0] = 0.8(0) + 5(2) = 10$$

Once the recursion has started, the remaining output values are easy to compute:

$$\begin{aligned} y[1] &= 0.8y[0] + 5x[1] = 0.8(10) + 5(-3) = -7 \\ y[2] &= 0.8y[1] + 5x[2] = 0.8(-7) + 5(0) = -5.6 \\ y[3] &= 0.8y[2] + 5x[3] = 0.8(-5.6) + 5(2) = 5.52 \\ y[4] &= 0.8y[3] + 5x[4] = 0.8(5.52) + 5(0) = 4.416 \\ y[5] &= 0.8y[4] + 5x[5] = 0.8(4.416) + 5(0) = 3.5328 \end{aligned}$$

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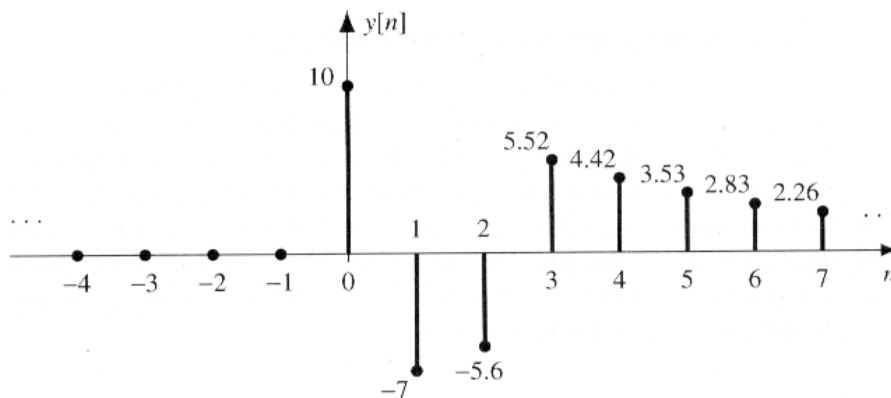


Figure 8.2. Output signal of the preceding recursive system.

It is important to note the structure of the output after the input signal has ended ($n > 3$). From that point onward, the difference equation becomes

$$y[n] = 0.8y[n - 1] \quad n > 3$$

and we can write the following closed-form expression

$$y[n] = y[3](0.8)^{n-3} \quad n \geq 3$$

8.2.1 Linearity and time invariance of IIR filters

The use of initial-rest conditions is sufficient to guarantee that a system implemented by iterating an IIR difference equation is both linear and time-invariant.

8.2.2 Impulse response of a first-order IIR filter

Because IIR systems with initial-rest conditions are LTI systems, the output can be expressed using the convolution sum:

$$y[n] = \sum_{k=-\infty}^{\infty} x[k]h[n-k]$$

To illustrate the characteristics of the impulse response $h[n]$ of an IIR filter, consider the following difference equation:

$$y[n] = a_1 y[n-1] + b_0 x[n]$$

By definition, the impulse response is

$$h[n] = a_1 h[n-1] + b_0 \delta[n]$$

From this, we can obtain the following general formula:

$$h[n] = \begin{cases} b_0(a_1)^n & \text{per } n \geq 0 \\ 0 & \text{per } n < 0 \end{cases}$$

which can be written as

$$h[n] = b_0(a_1)^n u[n]$$

where the product compactly expresses the conditions for $n < 0$ and for the indicated range. $(a_1)^n u[n]$

When the term $x[n-1]$ is also included in the difference equation,

$$y[n] = a_1 y[n-1] + b_0 x[n] + b_1 x[n-1]$$

because the system is linear and time-invariant, its impulse response can be expressed as the sum of two terms:

$$h[n] = b_0(a_1)^n u[n] + b_1(a_1)^{n-1} u[n-1]$$

The impulse response also decays exponentially at a rate determined only by the recursive coefficient. a_1

Worked example: impulse and step response. Consider the stable first-order system

$$y[n] = 0.8y[n-1] + 5x[n].$$

With initial-rest conditions, its impulse response is

$$h[n] = 5(0.8)^n u[n].$$

For a unit-step input, the output is a finite geometric sum: $y[n] = h[n] * x[n] \Leftrightarrow Y(z) = H(z)X(z)$

$$y[n] = 5 \sum_{k=0}^n (0.8)^k = 25(1 - (0.8)^{n+1})u[n].$$

The response approaches 25 because the pole magnitude is below one. This example connects recursion, the impulse response, the geometric-series formula, and the final steady value.

8.3 System function of an IIR filter

As in the FIR case, convolution in the n domain corresponds to multiplication in the z domain:

$$y[n] = h[n] * x[n] \Leftrightarrow Y(z) = H(z)X(z)$$

$$y[n] = a_1 y[n-1] + b_0 x[n] + b_1 x[n-1]$$

For FIR filters, the system function is a polynomial. When the difference equation is recursive, the system function is a ratio of two polynomials.

8.3.1 The general first-order case

The general form of a first-order difference equation is

$$y[n] = a_1 y[n-1] + b_0 x[n] + b_1 x[n-1]$$

Taking the z -transform of both sides of the equation gives

$$Y(z) = a_1 z^{-1} Y(z) + b_0 X(z) + b_1 z^{-1} X(z)$$

subtracting the output-transform term from both sides gives

$$Y(z) - a_1 z^{-1} Y(z) = b_0 X(z) + b_1 z^{-1} X(z)$$

$$(1 - a_1 z^{-1}) Y(z) = (b_0 + b_1 z^{-1}) X(z)$$

and, since the output transform can be factored, we can write

$$H(z) = \frac{Y(z)}{X(z)} = \frac{b_0 + b_1 z^{-1}}{1 - a_1 z^{-1}} = \frac{B(z)}{A(z)}$$

This shows that the system function $H(z)$ is the ratio of two polynomials. The numerator coefficients are the coefficients of the non-recursive part of the difference equation. The denominator has a leading coefficient equal to one, and its remaining coefficients are the negatives of the recursive coefficients.

The Python function `filter` uses the same convention. The code

```
yy = scipy.signal.lfilter(bb, aa, xx)
```

implements an IIR filter, where vectors `bb` and `aa` contain the numerator and denominator polynomial coefficients, respectively.

8.3.2 System function and block-diagram structures

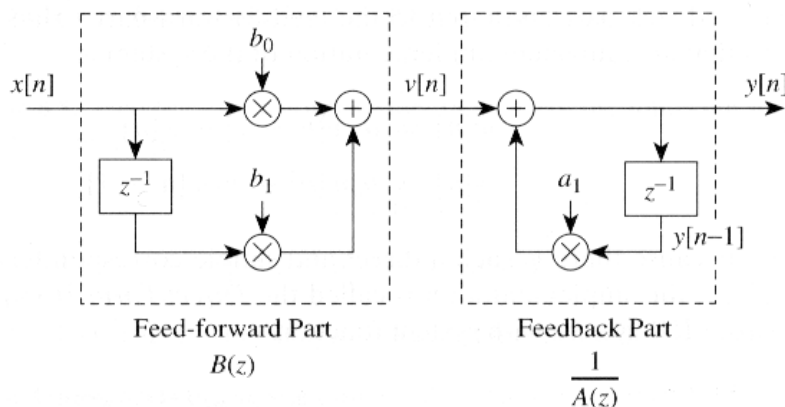


Figure 8.3. Direct-form I structure of an IIR filter.

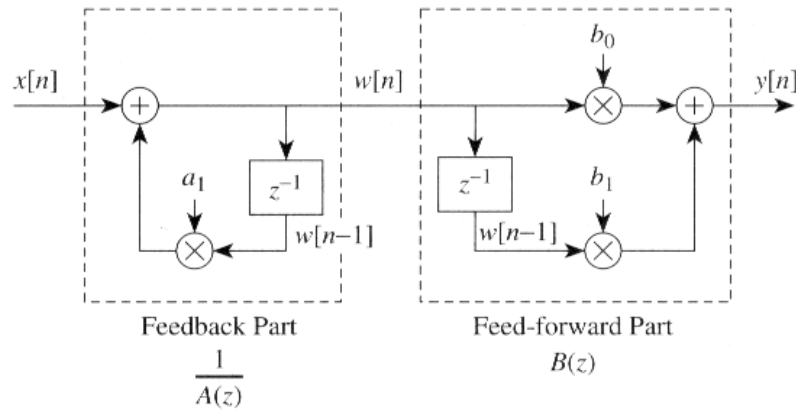


Figure 8.4. Direct-form II structure of an IIR filter.

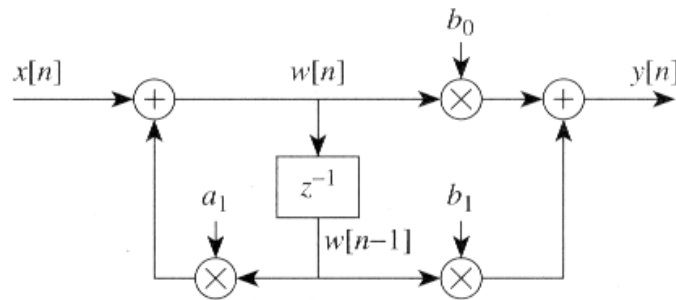


Figure 8.5. Direct-form II structure of an IIR filter with the delays combined.

Direct form II shows that two separate sets of delay units are unnecessary: they can be combined into a single delay chain, as shown in Figure 8.5.

8.3.3 Relation to the impulse response

So far, we have implicitly assumed that the system function $H(z)$ is the z -transform of the impulse response of an IIR system. This was already demonstrated for FIR filters, but for an IIR filter we must take the z -transform of an infinite-length sequence. As an example, consider an infinite exponential sequence. Applying the definition of the z -transform gives

$$H(z) = \sum_{n=0}^{\infty} a^n z^{-n} = \sum_{n=0}^{\infty} (az^{-1})^n$$

This is a geometric series in the indicated ratio. When the magnitude of the ratio is less than one, the sum is finite and can be written in closed form as

$$H(z) = \sum_{n=0}^{\infty} a^n z^{-n} = \frac{1}{1 - az^{-1}}$$

The condition under which the infinite sum can be expressed in closed form defines a set of z values in the complex plane called the region of convergence (ROC). From this analysis, we obtain the following transform pair:

$$a^n u[n] \longleftrightarrow \frac{1}{1 - az^{-1}}$$

As an example of this result, consider the difference equation

$$y[n] = a_1 y[n-1] + b_0 x[n] + b_1 x[n-1]$$

Its impulse response is

$$h[n] = b_0 (a_1)^n u[n] + b_1 (a_1)^{n-1} u[n-1]$$

Using linearity of the z-transform, the time-shift property, and the transform pair above, the system function is

$$H(z) = b_0 \left(\frac{1}{1 - a_1 z^{-1}} \right) + b_1 z^{-1} \left(\frac{1}{1 - a_1 z^{-1}} \right) = \frac{b_0 + b_1 z^{-1}}{1 - a_1 z^{-1}}$$

8.4 Poles and zeros

The roots of the numerator and denominator polynomials of $H(z)$ are central to the characterization of an IIR system. For a first-order system,

$$H(z) = \frac{b_0 + b_1 z^{-1}}{1 - a_1 z^{-1}} = \frac{b_0 z + b_1}{z - a_1}$$

the numerator root is

$$z = -\frac{b_1}{b_0}$$

and the denominator root is

$$z = a_1$$

A zero is a value of z for which $H(z)=0$. A pole is a value of z for which $H(z)$ becomes unbounded.

8.4.1 Zeros and poles at the origin or at infinity

When the numerator and denominator have different degrees, zeros or poles may occur at the origin or at infinity. Consider the recursive filter

$$y[n] = 0.5y[n-1] + 3x[n-1]$$

Its system function is

$$H(z) = \frac{3z^{-1}}{1 - 0.5z^{-1}} = \frac{3}{z - 0.5}$$

The pole is at $z=0.5$. Since

$$\lim_{z \rightarrow \infty} H(z) = 0$$

the system also has a zero at infinity. Counting zeros and poles at zero and infinity makes their total numbers equal.

8.4.2 Pole location and stability

The pole location determines the exponential behaviour of the impulse response. The first-order system function can be separated as

$$H(z) = b_0 \frac{1}{1 - a_1 z^{-1}} + b_1 z^{-1} \frac{1}{1 - a_1 z^{-1}}$$

and its impulse response is

$$h[n] = b_0 a_1^n u[n] + b_1 a_1^{n-1} u[n-1]$$

For a causal IIR system with initial-rest conditions, stability requires every pole to lie strictly inside the unit circle. In the first-order case, this condition is

$$|a_1| < 1$$

A useful special case is

$$H(z) = \frac{1}{1 - az^{-1}} \Leftrightarrow h[n] = a^n u[n]$$

Thus, $a=0.8$ produces a decaying stable response, whereas $a=1.1$ produces a growing unstable response. A pole close to the unit circle gives a long-lasting transient.

8.5 Frequency response of an IIR filter

The frequency response of a stable IIR system is obtained by evaluating its system function on the unit circle:

$$H(e^{j\hat{\omega}}) = H(z)|_{z=e^{j\hat{\omega}}}$$

For the general first-order system

$$H(z) = \frac{b_0 + b_1 z^{-1}}{1 - a_1 z^{-1}}$$

the region of convergence is

$$|a_1 z^{-1}| < 1 \Leftrightarrow |z| > |a_1|$$

Therefore, the unit circle belongs to the region of convergence only when $|a_1| < 1$. Under this stability condition,

$$H(e^{j\hat{\omega}}) = \frac{b_0 + b_1 e^{-j\hat{\omega}}}{1 - a_1 e^{-j\hat{\omega}}}$$

The response is periodic in normalized radian frequency with period 2π .

For possibly complex coefficients, its squared magnitude is

$$|H(e^{j\hat{\omega}})|^2 = \frac{|b_0|^2 + |b_1|^2 + 2\text{Re}\{b_0 b_1^* e^{j\hat{\omega}}\}}{1 + |a_1|^2 - 2\text{Re}\{a_1 e^{-j\hat{\omega}}\}}$$

When the coefficients are real, this simplifies to

$$|H(e^{j\hat{\omega}})|^2 = \frac{b_0^2 + b_1^2 + 2b_0 b_1 \cos \hat{\omega}}{1 + a_1^2 - 2a_1 \cos \hat{\omega}}$$

For real coefficients, the phase response can be written as

$$\angle H(e^{j\hat{\omega}}) = \tan^{-1} \left(\frac{-b_1 \sin \hat{\omega}}{b_0 + b_1 \cos \hat{\omega}} \right) - \tan^{-1} \left(\frac{a_1 \sin \hat{\omega}}{1 - a_1 \cos \hat{\omega}} \right)$$

Example. Consider

$$H(z) = \frac{2 + 2z^{-1}}{1 - 0.8z^{-1}}$$

This system has a pole at $z=0.8$ and a zero at $z=-1$. Hence its response is largest near zero frequency and vanishes at the Nyquist frequency:

$$H(e^{j0}) = H(1) = \frac{2 + 2}{1 - 0.8} = 20$$

$$H(e^{j\pi}) = H(-1) = 0$$

8.5.2 Three-dimensional representation of the system function

The relationship between the magnitude of $H(z)$ and the locations of its poles and zeros can be illustrated with three-dimensional plots. The frequency response consists of the values of $H(z)$ on the unit circle.

For example, consider the system function

$$H(z) = \frac{1}{1 - 0.8z^{-1}}$$

It can be illustrated by the following diagrams:

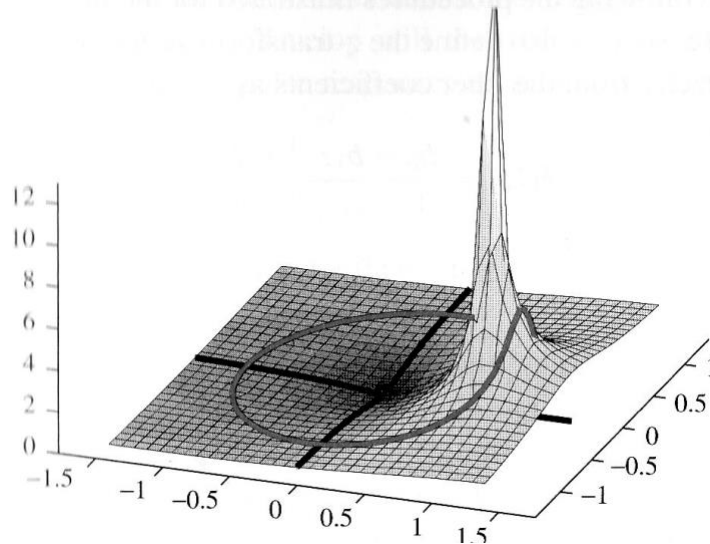


Figure 8.6. Z-transform magnitude of a first-order recursive filter. The pole is at $z=0.8$ and the numerator has a zero at $z=-1$.

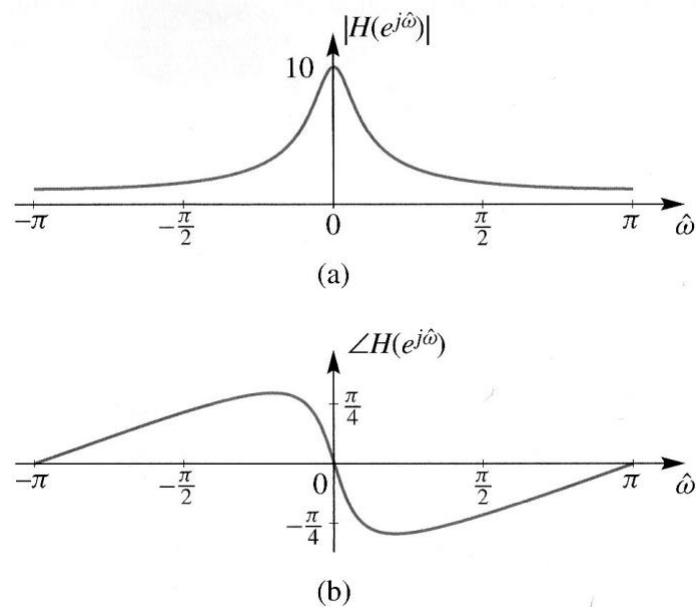


Figure 8.7. Magnitude and phase response of a first-order recursive filter. The pole is at $z=0.8$ and the numerator has a zero at $z=-1$.

8.6 The three domains

The difference equation provides the algorithm for computing the output signal from the input by recursively applying the filter coefficients. It also defines the impulse response.

The system function is a ratio of two polynomials, and its poles and zeros provide a compact set of parameters that completely characterizes the filter.

The shape of the frequency response, including passbands and stopbands, depends largely on the locations of the poles and zeros relative to the unit circle.

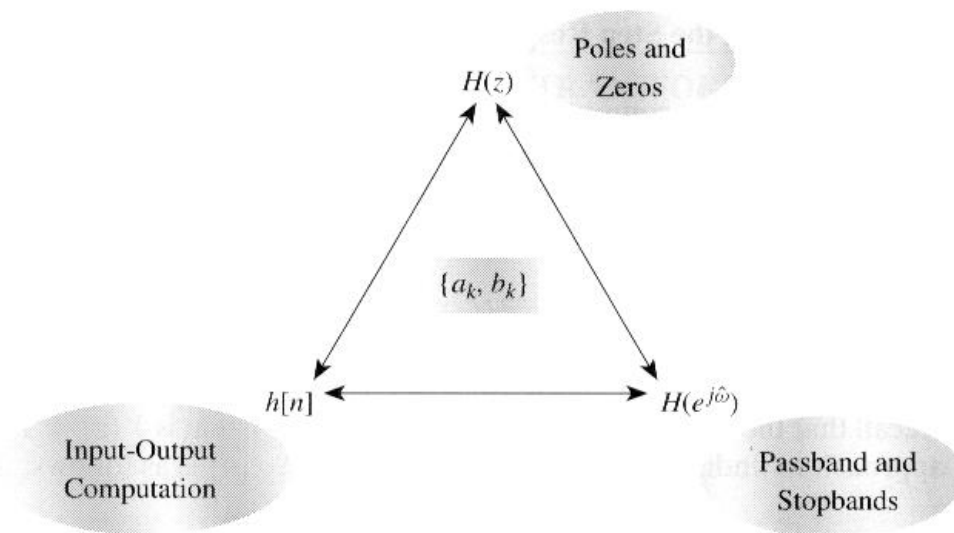


Figure 8.8. Relationship among the n , z , and frequency domains.

$x[n]$	$\leftrightarrow$	$X(z)$
$ax_1[n] + bx_2[n]$	$\leftrightarrow$	$aX_1(z) + bX_2(z)$
$x[n - n_0]$	$\leftrightarrow$	$z^{-n_0}X(z)$
$y[n] = x[n] * h[n]$	$\leftrightarrow$	$Y(z) = X(z)H(z)$
$\delta[n]$	$\leftrightarrow$	1
$\delta[n - n_0]$	$\leftrightarrow$	z^{-n_0}
$a^n u[n]$	$\leftrightarrow$	$\frac{1}{1 - az^{-1}}$

Table 8.1. Selected z -transform pairs.

8.7 The inverse z -transform and applications

The inverse z -transform converts a rational function in the z domain into its corresponding sequence in the n domain. For an LTI system,

$$Y(z) = H(z)X(z),$$

so the output can be obtained in three steps:

1. Compute $X(z)$, the z -transform of the input $x[n]$.
2. Form $Y(z) = H(z)X(z)$.
3. Compute the inverse z -transform of $Y(z)$ to obtain $y[n]$.

8.7.1 First-order system

Consider the first-order system function

$$H(z) = \frac{b_0 + b_1 z^{-1}}{1 - a_1 z^{-1}}.$$

For the unit-step input $x[n] = u[n]$,

$$X(z) = \frac{1}{1 - z^{-1}},$$

and therefore

$$Y(z) = H(z)X(z) = \frac{b_0 + b_1 z^{-1}}{(1 - a_1 z^{-1})(1 - z^{-1})}.$$

Assuming $a_1 \neq 1$, write the partial-fraction expansion as

$$Y(z) = \frac{A}{1 - a_1 z^{-1}} + \frac{B}{1 - z^{-1}}.$$

The residues are

$$A = Y(z)(1 - a_1 z^{-1})|_{z=a_1} = \frac{b_0 a_1 + b_1}{a_1 - 1},$$

$$B = Y(z)(1 - z^{-1})|_{z=1} = \frac{b_0 + b_1}{1 - a_1}.$$

Using the transform pair

$$\frac{1}{1 - p z^{-1}} \leftrightarrow p^n u[n],$$

the step response is

$$y[n] = A a_1^n u[n] + B u[n].$$

An equivalent form is

$$y[n] = \frac{b_0 + b_1 - (b_0 a_1 + b_1) a_1^n}{1 - a_1} u[n].$$

For the special case $b_1 = 0$,

$$y[n] = b_0 \frac{1 - a_1^{n+1}}{1 - a_1} u[n].$$

Worked example: inverse z-transform by partial fractions

Suppose

$$X(z) = \frac{1 - 2.1z^{-1}}{(1 + 0.5z^{-1})(1 - 0.8z^{-1})}.$$

Its partial-fraction expansion is

$$X(z) = \frac{2}{1 + 0.5z^{-1}} - \frac{1}{1 - 0.8z^{-1}},$$

so

$$x[n] = 2(-0.5)^n u[n] - (0.8)^n u[n].$$

8.7.2 General inverse z-transform procedure

Let $X(z)$ be a proper rational function with distinct poles $p_1, \dots, p_N$. Factor its denominator and write

$$X(z) = \sum_{k=1}^N \frac{A_k}{1 - p_k z^{-1}},$$

where

$$A_k = X(z)(1 - p_k z^{-1})|_{z=p_k}.$$

The corresponding causal sequence is

$$x[n] = \sum_{k=1}^N A_k p_k^n u[n].$$

When the numerator degree is not lower than the denominator degree, first perform polynomial division. Repeated poles require terms with higher powers of $(1 - p_k z^{-1})^{-1}$ and lead to polynomial factors in n multiplying p_k^n .

8.8 Steady-state response and stability

A system is bounded-input, bounded-output stable if every bounded input produces a bounded output. Consider

$$y[n] = a_1 y[n-1] + b_0 x[n],$$

with system function and impulse response

$$H(z) = \frac{b_0}{1 - a_1 z^{-1}},$$

$$h[n] = b_0 a_1^n u[n].$$

The causal system is stable precisely when

$$|a_1| < 1,$$

because this condition makes the impulse response absolutely summable and places the pole strictly inside the unit circle.

For a complex-exponential input that exists for all n ,

$$x[n] = e^{j\hat{\omega}_0 n},$$

the output is

$$y[n] = H(e^{j\hat{\omega}_0}) e^{j\hat{\omega}_0 n},$$

where

$$H(e^{j\hat{\omega}}) = \frac{b_0}{1 - a_1 e^{-j\hat{\omega}}}.$$

Now consider the suddenly applied input

$$x[n] = e^{j\hat{\omega}_0 n} u[n].$$

Its z-transform is

$$X(z) = \frac{1}{1 - e^{j\hat{\omega}_0} z^{-1}},$$

and hence

$$Y(z) = \frac{b_0}{(1 - a_1 z^{-1})(1 - e^{j\hat{\omega}_0} z^{-1})}.$$

Partial-fraction expansion gives

$$Y(z) = \frac{b_0 a_1}{a_1 - e^{j\hat{\omega}_0}} \frac{1}{1 - a_1 z^{-1}} + \frac{b_0}{1 - a_1 e^{-j\hat{\omega}_0}} \frac{1}{1 - e^{j\hat{\omega}_0} z^{-1}}.$$

Therefore,

$$y[n] = \frac{b_0 a_1}{a_1 - e^{j\hat{\omega}_0}} a_1^n u[n] + H(e^{j\hat{\omega}_0}) e^{j\hat{\omega}_0 n} u[n].$$

The first term is the **transient component**. It depends on the pole and decays when $|a_1| < 1$. The second is the **sinusoidal steady-state component**. For a stable system,

$$y[n] \rightarrow H(e^{j\hat{\omega}_0}) e^{j\hat{\omega}_0 n} \quad (n \rightarrow \infty).$$

If $|a_1| > 1$, the pole-dependent term grows and eventually dominates the output.

8.9 Second-order filters

A second-order IIR filter is described by

$$y[n] = a_1 y[n-1] + a_2 y[n-2] + b_0 x[n] + b_1 x[n-1] + b_2 x[n-2].$$

Taking the z-transform under initial-rest conditions gives

$$Y(z) = a_1 z^{-1} Y(z) + a_2 z^{-2} Y(z) + b_0 X(z) + b_1 z^{-1} X(z) + b_2 z^{-2} X(z).$$

Thus,

$$H(z) = \frac{Y(z)}{X(z)} = \frac{b_0 + b_1 z^{-1} + b_2 z^{-2}}{1 - a_1 z^{-1} - a_2 z^{-2}}.$$

If the denominator factors as

$$1 - a_1 z^{-1} - a_2 z^{-2} = (1 - p_1 z^{-1})(1 - p_2 z^{-1}),$$

then p_1 and p_2 are the poles. For real coefficients, the poles are either real or form a complex-conjugate pair.

8.9.1 Real and complex-conjugate poles

For complex-conjugate poles

$$p_1 = r e^{j\theta}, \quad p_2 = r e^{-j\theta},$$

the denominator becomes

$$1 - 2r \cos(\theta) z^{-1} + r^2 z^{-2}.$$

Therefore,

$$a_1 = 2r \cos(\theta), \quad a_2 = -r^2.$$

The pole radius r controls the decay or growth of the response, while the pole angle θ determines the oscillation frequency. The system is stable when both poles satisfy

$$|p_1| < 1, \quad |p_2| < 1.$$

8.9.2 Impulse response

For distinct poles, the system function can be expanded as

$$H(z) = C + \frac{A_1}{1 - p_1 z^{-1}} + \frac{A_2}{1 - p_2 z^{-1}},$$

where a direct term C is included when required by the numerator degree. The residues are

$$A_k = H(z)(1 - p_k z^{-1})|_{z=p_k}, \quad k = 1, 2.$$

The causal impulse response is

$$h[n] = C \delta[n] + A_1 p_1^n u[n] + A_2 p_2^n u[n].$$

When the poles are a complex-conjugate pair and the filter coefficients are real, the two exponential terms combine into a real damped sinusoid:

$$h[n] = Kr^n \cos(\theta n + \phi) u[n].$$

Worked example: second-order oscillator

Consider

$$H(z) = \frac{1}{1 - z^{-1} + z^{-2}}.$$

Its poles are

$$p_{1,2} = e^{\pm j\pi/3},$$

which lie on the unit circle. The impulse response is

$$h[n] = \frac{2}{\sqrt{3}} \sin\left((n+1)\frac{\pi}{3}\right) u[n].$$

Because the pole radius is one, the oscillation does not decay. Moving the poles to $re^{\pm j\pi/3}$ with $0 < r < 1$ produces a damped oscillation; placing them outside the unit circle produces an unstable growing oscillation.

8.10 Key points

- IIR filters are recursive: their outputs depend on both current or delayed inputs and previous outputs.
- Initial-rest conditions make the recursive realization causal, linear, and time invariant.
- The system function of an IIR filter is a rational function whose denominator roots are the poles.
- A causal IIR system is stable only when all poles lie strictly inside the unit circle.
- The frequency response is obtained by evaluating the system function on the unit circle, provided that the unit circle lies in the region of convergence.
- Inverse z-transforms and partial-fraction expansions convert rational functions into sums of causal exponential sequences.
- Complex-conjugate poles produce oscillatory responses; their radius controls decay and their angle controls oscillation frequency.

8.11 Notation

Symbol	Meaning
$x[n], y[n]$	Input and output discrete-time sequences
a_k	Recursive (feedback) filter coefficients
b_k	Non-recursive (feed-forward) filter coefficients
$h[n]$	Impulse response
$H(z)$	System function or transfer function
$X(z), Y(z)$	z-transforms of the input and output
z	Complex z-plane variable
ROC	Region of convergence of a z-transform
p_k, z_k	Pole and zero locations
$H(e^{j\omega})$	Frequency response evaluated on the unit circle
ω	Normalized radian frequency in radians per sample
$u[n], \delta[n]$	Unit-step and unit-impulse sequences
r, θ	Radius and angle of a complex-conjugate pole pair

